Design and Implementation of KARLS: A C++ KD-Tree-Based Robotic Location and Spatial Intelligence System with Applications in AI, Quantum Computing, and Blockchain.

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Course: Data Structures and Algorithms II | BLAST 02

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Abstract

This project presents the **design and implementation of KARLS (KD-Tree Assisted Robotic Location System)**, a **C++-based robotic location and spatial intelligence application** developed to support efficient **object-location management and nearest-neighbour searching** within a simulated warehouse environment. The primary goal is to demonstrate how **spatial data structures** can organize object coordinates, identify nearby objects, and support **location-based robotic decision-making**, providing a foundation for future applications in **Artificial Intelligence (AI) and Robotics**.

KARLS incorporates core **Data Structures and Algorithms (DS II)** concepts, including **two-dimensional KD-Trees, binary trees, recursion, pointers, dynamic memory allocation, and Object-Oriented Programming (OOP)**. Its **menu-driven interface** allows users to add objects, search by name, update the robot's position, identify the nearest object, calculate distances, display tree structures, and remove stored objects. **UML class diagrams** represent the application's design and functional components.

The system organizes objects through **alternating X-axis and Y-axis comparisons** and applies **Euclidean distance calculations** with **recursive nearest-neighbor searching and branch pruning**. For a reasonably balanced KD-Tree, insertion typically requires **O(log n)** time, with **O(n)** worst-case performance. Nearest-neighbor searching can reduce unnecessary comparisons compared with **O(n) linear search**, although performance depends on **tree balance, spatial distribution, and object count** and may also reach **O(n)** in the worst case.

In a practical warehouse simulation, KARLS identifies the nearest stored object as the robot's coordinates change. The application **manages and searches simulated locations** but does not physically navigate or calculate collision-free paths.

This project demonstrates the significance of **spatial indexing, algorithmic efficiency, modular programming, and object-oriented design**, establishing a foundation for future research involving **AI, Quantum Computing, Blockchain Technology, and autonomous robotic systems.**

Future Work

Artificial Intelligence (AI) and Machine Learning (ML): Future enhancements may incorporate **intelligent object recognition, computer vision, predictive spatial analysis, and automated robotic decision-making** to improve object identification and location-based task management.

Quantum Computing: Quantum-inspired optimization and quantum search research may be investigated for **complex spatial searching, multi-robot coordination, route optimization, and robotic task scheduling**, with performance evaluated against conventional algorithms.

Blockchain Technology: Blockchain and cryptographic verification could be explored to support **secure location-data sharing, tamper-evident event histories, trusted robotic transactions, and coordination among multiple autonomous robots**.

Autonomous Robotics: Future improvements may integrate **LiDAR sensors, real-time obstacle detection, autonomous navigation, collision-free path planning, automatic KD-Tree balancing, interactive warehouse mapping, cloud synchronization, and persistent data storage** to develop a more **intelligent, secure, and scalable robotic location-management system**.